Ifrat Ikhtear Uddin

312-843-1570 | ifrat.ikhtear@gmail.com | Google Scholar | GitHub | LinkedIn | Website

Technical Skills

AI / Machine Learning: PyTorch, Keras, Explainable AI (XAI), Adversarial Machine Learning, Deep Learning
Programming Languages: Python, JavaScript, TypeScript, Java
Frontend Technologies: Angular, React.js, Next.js, Redux, PrimeNG, HTML5, CSS3, SCSS
Backend & APIs: Node.js, Spring Boot, RESTful APIs, SQL
DevOps & Tools: Git, CI/CD, Docker, Kubernetes, WCAG 2.2 AA

Education

- **PhD in Data Science & Engineering** *August 2026 - May 2028*
  University of South Dakota, SD, USA

- **Master of Science in Computer Science** *January 2025 - May 2026*
  University of South Dakota, SD, USA GPA: 4.00

Publications

- Wang, L., **Uddin, I. I.**, Santosh, K. C., Zhang, C., Qin, X., & Zhou, Y. Bridging Symmetry and Robustness: On the Role of Equivariance in Enhancing Adversarial Robustness. In the Proceedings of NeurIPS 2025 [**Link**]
- Wang, L., **Uddin, I. I.**, Qin, X., Zhou, Y., & Santosh, K. C. (2025). Explainability-driven defense: Grad-CAM-guided model refinement against adversarial threats. In Proceedings of the AAAI 2025 Summer Symposium [**Link**]
- **Uddin, I. I.**, Wang, L., & Santosh, K. C. (2025). Expert-Guided Explainable Few-Shot Learning for Medical Image Diagnosis. *MICCAI Workshop on Data Engineering in Medical Imaging (DEMI)*, LNCS vol. 16191, Springer. [**Link**]
- Wang, L., **Uddin, I. I.**, & Santosh, K. C. (2026). Expert-Guided Explainable Few-Shot Learning with Active Sample Selection for Medical Image Analysis. IEEE Journal of Biomedical and Health Informatics [**Link**]
- **Uddin, I. I.**., & Zhou, Y., Zhang, C., Wang L. Explainable Category Discovery on Semantic Concept Space [**Accepted IEEE COGMI 2026**]

Under Review / In Preparation

  - **Uddin, I. I.**., M Mozafari-Nia, Wang L., & C Chen. Discovering What and How Many: Interpretable Generalized Category Discovery with Dirichlet Process Mixtures [**In Progress**]

Experience

- **Graduate Research Assistant** *June 2025 – Present*
  University of South Dakota SD, USA

  – Lead a research line on **explainable open-world recognition**: concept-bottleneck models that discover novel categories in unlabeled data, estimate how many there are with Bayesian nonparametrics, and explain every prediction and discovered class through named concepts; one first-author manuscript under review and a second in preparation, in collaboration with the University of Maine.
  – Co-authored four peer-reviewed papers (NeurIPS, IEEE JBHI, AAAI Symposium etc) on explanation-guided learning and architecture-level adversarial robustness within 18 months of entering ML research.
  – Build and benchmark end-to-end experimental pipelines on real-world datasets to validate trustworthy AI systems through explainability and adversarial robustness evaluation.
  – Serve as a reviewer for ML journals and conferences including IEEE Transanction in medical imaging, ICML Workshops etc.
  – Tech Stack: Python, PyTorch

- **Software Engineer** *March 2023 - December 2024*
  Kaz Software Limited Dhaka, Bangladesh

  – Contributed to the design and development of IBFD's Tax Research Platform, collaborating with international teams to implement new features, refactor legacy code, resolve technical challenges and in addition worked on improving platforms accessibility in compliance with Web Content Accessibility Guidelines (WCAG) 2.2 AA standards.

- Worked closely with back-end developers to design and develop RESTful APIs for front-end integration, leveraging a full architectural understanding of the system.
- Worked as a Front-End Software Engineer using Angular; led junior developers, ensured end-to-end application maintenance, implemented new features, and conducted R&D for feature enhancements.
- Tech Stack: Angular, TypeScript, JavaScript, Java

• **Software Engineer** *July 2022 - Feb 2023*

Brain Station 23 PLC Dhaka, Bangladesh

- Designed and developed three Angular-based client applications, including two for a financial platform.
- Collaborated with back-end engineers to design and integrate APIs for seamless front-end functionality.
- Supported multiple live projects by fixing bugs, optimizing code, and implementing new features.
- Mentored junior developers on front-end technologies and software development best practices.
- Tech Stack: Angular, .NET, Oracle, SQL

• **Junior Software Engineer** *June 2020 - June 2022*

Square Health Limited Dhaka, Bangladesh

- Collaborated with cross-functional software engineering teams to develop new features and resolve existing system issues.
- Led the development of key functionalities, including audio/video calling, dental module integration, medicine suggestion settings, and dynamic prescription design.
- Conducted research and feasibility analysis to identify optimal implementation strategies for upcoming features.
- Evaluated and integrated tools, libraries, APIs, and frameworks to enhance performance and ensure security in a data-intensive web environment.
- Tech Stack: JavaScript, TypeScript, Angular, Dexie.js, Moment.js

Projects

• **xGCD: Explainable Generalized Category Discovery** *Ongoing*

Interpretable discovery of an unknown number of novel classes Lead researcher

- Designed a three-stage GCD framework that runs entirely in a explainable concept space, so every cluster assignment and the estimated class count are explained by named visual attributes.
- Estimate the number of novel classes instead of assuming it: on the frozen concept logits, a tied-covariance (LDA) Gaussian model with a chi-squared-motivated, data-calibrated Mahalanobis novelty test routes samples to a novel pool; a Dirichlet Process mixture with CRP prior counts the classes; cluster-level absorption/merge and a label-free concept-coherence gate clean the partition.
- Deliberately separate counting from refinement: the count is fixed on the frozen representation, then a prototype-initialised linear head is refined with confidence-weighted pseudo-labels and a concept-fidelity anchor, avoiding the class collapse of joint EM approaches.
- Grow the concept vocabulary for novel classes proposes concepts from each cluster's representative images, enrichment-ratio filtering drops non-specific ones, and the CBL is expanded with known-concept parameters frozen.

• **xNCD: Explainable Novel Category Discovery in Concept Space** *Under review*

Concept-bottleneck model for novel category discovery First author

- Proposed a framework that unifies multi-view self-labelling for category discovery with a label-free, CLIP-aligned concept bottleneck, so that each discovered cluster is explained by named semantic concepts rather than opaque features.
- Established CLIP concept alignment as the interpretability guarantee of the method and benchmarked against NCD/GCD baselines on standard benchmarks.

• **Tax Research Platform [Live Link]**

A subscription-based international tax research platform developed by IBFD, Netherlands.

- Implemented new features and optimized existing modules through code refactoring.
- Conducted R&D on transitioning the monolithic structure to a micro front-end architecture.
- **Tech Stack:** Angular, PrimeNG, AmCharts, Java

• **Jotno [Live Link]**

A telemedicine platform used by thousands of doctors in Bangladesh, developed under Square Health.

- Led front-end development for both doctor and patient portals.
- Developed key modules including audio/video calling, dynamic prescription templates, and medicine suggestion settings.
- **Tech Stack:** Angular, PrimeNG, Dexie.js

Volunteering/Open-Source Contribution

- **Prime NG:** Contributed to the open-source project Prime NG, a widely used UI Component Library for frontend development. Provided solutions to various issues listed on their GitHub issue board, enhancing the library's functionality and user experience.

  –GitHub: Prime NG Repository

- **WirHelfen.eu:** Collaborated with WirHelfen.eu, a non-profit humanitarian platform focused on aiding individuals in need. Contributed to the design and development of a responsive and user-friendly UI to assist people in accessing support

  –Website: WirHelfen.eu